Cardiology I: Ischemic Heart Disease (Chronic Coronary Disease and ACS)

All key points for the exam (ACC/AHA 2023 Chronic Coronary Disease, 2025 ACS guideline, ESC) — drug tables included

1. Pathophysiology and Classification

- **Ischemic heart disease (IHD) / coronary artery disease (CAD)** = imbalance between **myocardial O2 supply** (coronary blood flow — atherosclerotic narrowing, thrombosis, vasospasm) and **O2 demand** (determined by **heart rate**, **contractility**, and **wall tension** — preload/afterload).
- **Heart rate is the most important determinant of demand** • coronary perfusion occurs mostly in **diastole** → slowing HR ↑ supply and ↓ demand.
- **Atherosclerosis:** endothelial dysfunction → LDL accumulation/oxidation → foam cells → fatty streak → fibrous plaque • **plaque rupture/erosion + thrombosis** → ACS.

Syndrome	Features
Chronic coronary disease (CCD) / stable ischemic heart disease	**Stable angina**: predictable chest pain with **exertion or emotional stress**, lasts < 10–15 min, relieved by **rest or SL nitroglycerin** • fixed stenosis (usually ≥ 70%)
Unstable angina (UA)	New-onset, rest, or **crescendo** (more frequent, longer, less exertion) angina • **troponin negative** • part of NSTE-ACS
NSTEMI	Ischemic symptoms + **↑ troponin** (myocardial necrosis) • ECG: ST depression, T-wave inversion, or normal • partial/intermittent occlusion
STEMI	**ST elevation** in ≥ 2 contiguous leads (or new LBBB with ischemic symptoms/posterior MI) + ↑ troponin • **complete occlusion** → emergent reperfusion
Variant (Prinzmetal, vasospastic) angina	**Rest angina** (often early morning) due to **coronary vasospasm** • transient ST elevation • smokers, cocaine, triptans • treat with **CCBs + nitrates** • **avoid nonselective β-blockers** (unopposed α → worsen spasm)
Microvascular angina / INOCA / MINOCA	Ischemia with no obstructive CAD (women) • β-blockers, CCBs, ACEi, statins

- **CCS (Canadian Cardiovascular Society) angina classes:** I (angina only with strenuous exertion) → II (slight limitation) → III (marked limitation, walking 1–2 blocks) → IV (angina at rest/any activity).
- **Risk factors: non-modifiable** — age (men ≥ 45, women ≥ 55), male sex, family history of premature CAD • **modifiable** — **dyslipidemia**, **hypertension**, **diabetes**, **smoking**, obesity, physical inactivity, poor diet, CKD, chronic inflammatory disease, stress.
- **Atypical presentations** (no chest pain — dyspnea, fatigue, nausea, epigastric discomfort): **women**, **elderly**, **diabetics** (autonomic neuropathy).

2. Diagnosis of CAD

- **12-lead ECG** (at rest often normal in stable angina) • **stress testing**: exercise ECG, stress echo, nuclear (SPECT/PET), stress cardiac MRI • pharmacologic stress if unable to exercise: **vasodilators** (**adenosine**, **regadenoson**, dipyridamole) for nuclear imaging or **dobutamine** for echo.
- **Hold before vasodilator stress: caffeine and methylxanthines (theophylline) 12–24 h** (adenosine receptor antagonists) • dipyridamole ↑ adenosine effect • hold β-blockers/nitrates/CCBs before exercise tests if diagnostic (per protocol).
- **Coronary CT angiography (CCTA)** • **coronary artery calcium (CAC)** score • **invasive coronary angiography** (gold standard) + FFR.
- **Troponin** (high-sensitivity) = biomarker of choice for MI • rises 2–3 h, peaks ~24 h, stays elevated 7–14 days • serial measurements (0/1 h or 0/2 h algorithms) • CK-MB rarely used now.
- **Echocardiogram** for LV function (LVEF) — guides therapy (β-blocker, ACEi, MRA, ICD).

3. Chronic Coronary Disease (CCD) — ACC/AHA 2023

Goals: (1) **reduce risk of MI and death** (event-reducing therapy) and (2) **relieve angina** and improve quality of life (antianginal therapy).

Lifestyle and risk-factor management

- **Smoking cessation** (varenicline, NRT, bupropion) • **heart-healthy diet** (Mediterranean/DASH) • **physical activity** ≥ 150 min/week moderate • **weight loss** if overweight • limit alcohol • **cardiac rehabilitation** • manage stress/depression • **annual influenza vaccine** (also COVID, pneumococcal).
- **BP goal < 130/80 mmHg • diabetes:** A1C individualized (< 7%) + **SGLT2 inhibitor and/or GLP-1 RA** with proven CV benefit regardless of A1C.

Event-reducing ("secondary prevention") pharmacotherapy

Therapy	Recommendation (CCD)
Antiplatelet	**Aspirin 75–100 mg daily** (lifelong) • **clopidogrel 75 mg** if aspirin intolerant/allergic (reasonable alternative to aspirin) • **DAPT after elective PCI: 6 months** (clopidogrel + aspirin) → then single antiplatelet; 1–3 months if high bleeding risk • selected high ischemic risk: longer DAPT or **rivaroxaban 2.5 mg BID + aspirin** (COMPASS) • routine DAPT without PCI not recommended
Statin	**High-intensity statin** (atorvastatin 40–80 mg, rosuvastatin 20–40 mg) → goal **≥ 50% LDL reduction** • **LDL < 70 mg/dL** reasonable; very high risk → **LDL < 55** • add **ezetimibe** → then **PCSK9 inhibitor** (or inclisiran/bempedoic acid) if not at goal
β-blocker	Recommended if **LVEF ≤ 50%** (evidence-based HF β-blocker if ≤ 40%) • after MI: reasonable for 1 year • **long-term β-blocker NOT needed** for event reduction if no MI within past year, LVEF > 50%, and no other indication (angina, arrhythmia, HTN) → may reassess/stop
ACEi / ARB	If **HTN, diabetes, CKD, or LVEF ≤ 40%** (also reasonable in others) • ARNI if HFrEF
SGLT2i / GLP-1 RA	**Type 2 diabetes** (CV benefit independent of A1C) • SGLT2i also for HF/CKD • GLP-1 RA (semaglutide) for obesity with CCD (↓ MACE — SELECT)
Colchicine	**0.5 mg daily** may be considered to ↓ recurrent ASCVD events (LoDoCo2) • avoid in severe CKD/hepatic disease • DDIs: strong CYP3A4/P-gp inhibitors
MRA	LVEF ≤ 40% with HF or diabetes after MI
Not recommended	**Hormone replacement therapy**, vitamins (C, E, β-carotene), fish oil supplements (except icosapent ethyl), chelation, NSAIDs (↑ CV events)

Antianginal therapy

- **All patients: sublingual nitroglycerin** (or spray) for acute angina relief and prophylaxis before exertion.
- **First-line maintenance: β-blocker or CCB** (choice by comorbidities) • **add or substitute: long-acting nitrate**, **ranolazine**, or the other class (β-blocker + DHP CCB) • **avoid β-blocker + non-DHP CCB** (verapamil/diltiazem) → bradycardia/AV block/HF (use cautiously only if needed).
- **Revascularization** (PCI or CABG) for refractory angina despite GDMT, or to improve survival in high-risk anatomy (left main, multivessel with ↓ LVEF/diabetes → CABG).

Class	Mechanism / effect	Key points
β-blockers (metoprolol, atenolol, bisoprolol, carvedilol)	↓ HR, ↓ contractility → **↓ O2 demand** • ↑ diastolic filling time	**Target resting HR 55–60 bpm** • prefer **β1-selective without ISA** (avoid pindolol/acebutolol — ISA) • ↓ mortality post-MI and in HFrEF • ADRs: bradycardia, fatigue, bronchospasm (non-selective), masked hypoglycemia, sexual dysfunction, depression • **don't stop abruptly** (rebound angina/MI) — taper over 1–2 weeks • CI: severe bradycardia, high-grade AV block, decompensated HF, cardiogenic shock • **avoid in vasospastic angina** and cocaine-induced ACS

Class	Mechanism / effect	Key points
CCBs — DHP (amlodipine, felodipine, nifedipine ER)	Coronary and peripheral vasodilation → ↓ afterload, ↑ supply (prevent spasm)	**Avoid short-acting (IR) nifedipine** (reflex tachycardia → ↑ MI) • ADRs: **peripheral edema**, headache, flushing, reflex tachycardia, gingival hyperplasia • amlodipine/felodipine safe in HFrEF
CCBs — non-DHP (verapamil, diltiazem)	↓ HR, ↓ contractility, coronary dilation	**Avoid in HFrEF** (negative inotropes) and with β-blockers • ADRs: bradycardia, AV block, **constipation (verapamil)**, edema • **CYP3A4 inhibitors** (↑ simvastatin, lovastatin)
Nitrates (NTG SL/spray/patch, isosorbide mononitrate, isosorbide dinitrate)	Release **nitric oxide** → venodilation → **↓ preload** (↓ O2 demand) + coronary dilation	**Tolerance** → **nitrate-free interval 10–14 h/day** • ADRs: **headache** (most common), **hypotension**, dizziness, flushing, **reflex tachycardia** (add β-blocker) • **CI with PDE-5 inhibitors** (sildenafil, vardenafil within 24 h; **tadalafil within 48 h**) and **riociguat** → severe hypotension • CI in RV infarction, severe aortic stenosis
Ranolazine (Ranexa)	Inhibits **late sodium current** → ↓ intracellular Ca → ↓ wall tension • **no effect on HR or BP**	500 mg BID → max **1,000 mg BID** (ER, don't crush) • ADRs: dizziness, constipation, nausea, headache, **QT prolongation** (dose-related) • **CYP3A4 substrate** → **CI with strong 3A4 inhibitors** (ketoconazole, clarithromycin, ritonavir) and **inducers** (rifampin, carbamazepine, St John's wort); max 500 mg BID with moderate inhibitors (diltiazem, verapamil, fluconazole) • P-gp inhibitor → ↑ digoxin • limit **simvastatin to 20 mg**, **metformin to 1,000 mg/day** • CI in **liver cirrhosis** • useful when BP/HR low
Ivabradine	I(f) "funny" current inhibitor (SA node) → ↓ HR	Approved for HFrEF (not for angina in US); used for angina in Europe

Nitroglycerin — patient counseling (very high-yield)

- **Sublingual NTG 0.4 mg** (tablet or spray) at onset of chest pain — **sit down** (dizziness) • **if pain not improved or worse 5 minutes after the first dose → call 911 immediately** (2025 counseling), then may take a 2nd dose; may repeat every 5 min up to **3 doses** total.
- Can be used **5–10 min before activities** that cause angina (prophylaxis).
- Tablets: keep in the **original glass amber container**, tightly closed, away from heat/moisture (not in pill boxes or the car); should cause slight burning/tingling (not reliable sign of potency) • spray: don't shake, don't inhale, prime before first use.
- **Don't use with PDE-5 inhibitors** (sildenafil/vardenafil 24 h, tadalafil 48 h) • headache is common (acetaminophen).
- **Transdermal NTG patch** (0.2–0.8 mg/h): apply in the morning, **remove after 12–14 h** (patch-free interval at night) • **isosorbide mononitrate ER** 30–60 mg → max 240 mg **once daily in the morning** • IR ISMN 20 mg BID 7 h apart (8 AM and 3 PM) • **isosorbide dinitrate** 10–40 mg TID (e.g., 8 AM, 1 PM, 6 PM — 14-h gap).

4. Acute Coronary Syndromes (ACS) — Diagnosis and Initial Management

- **Symptoms: chest pain/pressure/tightness** (> 20 min, at rest) radiating to the **left arm, jaw, neck, back**, with **dyspnea**, **diaphoresis**, nausea, lightheadedness • atypical in women, elderly, diabetics.
- **ECG within 10 minutes** of arrival → **STEMI** (ST elevation) vs **NSTE-ACS** (NSTEMI/UA) • serial hs-troponin.
- **Risk stratification (NSTE-ACS): GRACE** (mortality) and **TIMI** scores → guide timing of invasive strategy.
- **Complications of MI:** arrhythmias (VF — sudden death, AF, heart block), **heart failure/cardiogenic shock**, mechanical (papillary muscle rupture → acute MR, ventricular septal rupture, free wall rupture/tamponade), LV thrombus, pericarditis (early; Dressler syndrome later), recurrent ischemia.

Immediate ("early") pharmacotherapy for all ACS

Therapy	Dose / details
Aspirin	**162–325 mg non-enteric-coated, chewed** immediately (as soon as possible) → **81 mg daily** indefinitely (**≤ 100 mg/day with ticagrelor** — higher doses ↓ ticagrelor efficacy) • allergy → clopidogrel (or desensitization)
Nitroglycerin	**SL 0.4 mg every 5 min × 3** for ongoing ischemic pain → **IV NTG** (5–10 mcg/min, titrate) for persistent pain, **HTN**, or **HF/pulmonary congestion** • **CI:** SBP < 90 or ≥ 30 mmHg below baseline, **RV infarction** (inferior MI — preload dependent), HR < 50 or > 100 (without HF), **PDE-5 inhibitor use** (24–48 h) • symptom relief only (no mortality benefit)
Oxygen	Only if **SpO2 < 90%**, respiratory distress, or high-risk hypoxemia (routine O2 not beneficial)
Morphine	2–4 mg IV for pain **unresponsive** to nitrates • **caution**: may ↓ absorption/onset of oral P2Y12 inhibitors (opioid slows GI motility), hypotension, respiratory depression
P2Y12 inhibitor	Loading dose (see table) — timing depends on strategy (at PCI or at presentation)
Parenteral anticoagulant	**UFH**, **enoxaparin**, **bivalirudin**, or fondaparinux (see below)
High-intensity statin	**Atorvastatin 80 mg** (or rosuvastatin 20–40 mg) **as early as possible**, regardless of baseline LDL
β-blocker	**Oral** β-blocker **within 24 h** if no contraindications: signs of **HF**, **low-output state**, **risk of cardiogenic shock** (age > 70, SBP < 120, HR > 110 or < 60, late presentation), **PR > 0.24 s**, 2nd/3rd-degree AV block, active asthma/reactive airway disease • strongest evidence with **LVEF ≤ 40%** • **IV β-blocker** only for HTN/ongoing ischemia without contraindications (↑ cardiogenic shock — COMMIT) • **avoid in cocaine-induced ACS**
ACEi (or ARB)	**Within 24 h** if **anterior MI**, **LVEF ≤ 40%**, **HF**, HTN, diabetes, or CKD (and no hypotension) • oral only (**no IV enalaprilat**)
Avoid	**NSAIDs** (except aspirin) — ↑ reinfarction, HF, rupture, death • **IV β-blockers** in at-risk patients • **immediate-release nifedipine** • **fibrinolytics in NSTE-ACS** • routine O2

Mnemonic for early ACS therapy: "MONA-BASH" (modernized)

- **M**orphine (if refractory pain), **O**xygen (only if SpO2 < 90%), **N**itroglycerin, **A**spirin • **B**eta-blocker, **A**CE inhibitor, **S**tatin, **H**eparin (anticoagulant) • + P2Y12 inhibitor • + reperfusion.

5. STEMI — Reperfusion

- **Primary PCI is preferred** if it can be performed within **120 min of first medical contact (FMC)** • goals: **FMC-to-device ≤ 90 min** (PCI-capable hospital) or **≤ 120 min** (transfer) • symptoms **< 12 h** (PCI also reasonable 12–24 h with ongoing ischemia, any time for cardiogenic shock).
- **Fibrinolysis** if PCI **not available within 120 min** and symptoms **< 12 h** → **door-to-needle ≤ 30 min** • then **transfer to PCI-capable center** — **immediate angiography if failed reperfusion** (rescue PCI), routine angiography **within 3–24 h** after successful lysis (pharmaco-invasive).
- **Signs of successful reperfusion:** ≥ 50% ST resolution at 60–90 min, symptom relief, reperfusion arrhythmias (accelerated idioventricular rhythm).
- **Fibrinolytics are NOT indicated in NSTE-ACS** (harm).

Fibrinolytic	Dose	Notes
Tenecteplase (TNK)	**Single IV bolus over 5 s, weight-based** (30–50 mg: < 60 kg 30 mg → ≥ 90 kg 50 mg; 0.5 mg/kg) • **half-dose in age ≥ 75** (STREAM-2)	Most fibrin-specific • easiest (single bolus) • preferred
Alteplase (tPA)	**15 mg IV bolus → 0.75 mg/kg over 30 min (max 50 mg) → 0.5 mg/kg over 60 min (max 35 mg)** (max total 100 mg)	Accelerated 90-min infusion
Reteplase	**10 units IV bolus × 2**, 30 min apart	Double bolus
Streptokinase	1.5 million units over 60 min	Not fibrin-specific, antigenic (allergy), hypotension • not available in US

- **Adjunctive therapy with fibrinolysis: aspirin** 162–325 mg + **clopidogrel 300 mg load** (age ≤ 75) → 75 mg daily • **age > 75: clopidogrel 75 mg, NO loading dose** • (prasugrel/ticagrelor not studied with lysis; ticagrelor may be switched to after 48 h) • **anticoagulant** for ≥ 48 h up to 8 days or until revascularization: **enoxaparin preferred** (30 mg IV bolus + 1 mg/kg SC q12h; **age ≥ 75: no bolus, 0.75 mg/kg q12h**; CrCl < 30: 1 mg/kg daily) or UFH (60 units/kg bolus max 4,000 → 12 units/kg/h max 1,000; aPTT 1.5–2×) or fondaparinux.
- **Main risk: bleeding — intracranial hemorrhage (~1%)** (↑ age, female, low weight, HTN, prior stroke).

Absolute contraindications to fibrinolysis (STEMI)	Relative contraindications
• **Any prior intracranial hemorrhage** • Known **structural cerebral vascular lesion** (AVM) or **malignant intracranial neoplasm** • **Ischemic stroke within 3 months** (except acute stroke within 4.5 h) • Suspected **aortic dissection** • **Active bleeding** or bleeding diathesis (excluding menses) • **Significant closed-head or facial trauma within 3 months** • Intracranial or intraspinal surgery within 2 months • Severe uncontrolled HTN unresponsive to emergency therapy • For streptokinase: prior use within 6 months	• Chronic, severe, poorly controlled HTN • **SBP > 180 or DBP > 110** on presentation • Ischemic stroke > 3 months, dementia • Traumatic or prolonged (> 10 min) CPR • **Major surgery < 3 weeks** • Recent internal bleeding (2–4 weeks) • Noncompressible vascular punctures • **Pregnancy** • Active peptic ulcer • **Oral anticoagulant therapy**

6. NSTE-ACS (Unstable Angina / NSTEMI) — Invasive Strategy

- **Immediate invasive (< 2 h): refractory angina**, **hemodynamic instability/cardiogenic shock**, **acute HF**, sustained VT/VF, mechanical complications.
- **Early invasive (< 24 h):** high risk — **GRACE > 140**, dynamic ST changes, rise/fall of troponin consistent with NSTEMI.
- **Invasive during hospitalization** (intermediate risk) or **selective/ischemia-guided** (low risk — stress test first).
- All receive aspirin, P2Y12 inhibitor, anticoagulant, high-intensity statin, β-blocker (if no CI), ± ACEi • **no fibrinolytics**.

7. Antiplatelet and Anticoagulant Therapy in ACS

P2Y12 inhibitors

Drug	Loading → maintenance	Key points
Clopidogrel (Plavix)	**600 mg** (PCI) or **300 mg** (fibrinolysis; **no load if > 75 years**) → **75 mg daily**	**Prodrug** activated by **CYP2C19** → **boxed warning: poor metabolizers** (↓ effect) → genotype-guided alternative • **avoid omeprazole/esomeprazole** (2C19 inhibitors) → use **pantoprazole** • irreversible • **hold 5 days before CABG** • used with fibrinolysis, OAC (triple therapy), high bleeding risk, CCD • TTP (rare)
Prasugrel (Effient)	**60 mg** → **10 mg daily** (**5 mg if < 60 kg**)	Prodrug (not 2C19-dependent) — more potent, faster • **only in patients undergoing PCI** (after anatomy known) • **boxed: CONTRAINDICATED with prior stroke/TIA** • **not recommended age ≥ 75** (except high-risk diabetes/prior MI) • ↑ bleeding • **hold 7 days before CABG**
Ticagrelor (Brilinta)	**180 mg** → **90 mg BID** × 12 months → **60 mg BID** (extended, up to 3 years after MI — PEGASUS)	**Reversible**, not a prodrug • used in invasive or conservative strategy • **boxed: aspirin dose must be ≤ 100 mg/day** • **dyspnea** (~15%, adenosine-mediated), **bradyarrhythmias** (ventricular pauses), ↑ uric acid, ↑ SCr • **CYP3A4 substrate**: avoid strong inhibitors (ketoconazole, clarithromycin, ritonavir) and inducers (rifampin) • ↑ simvastatin/lovastatin (max 40 mg) • ↑ digoxin • **hold 3–5 days before CABG** • CI prior ICH
Cangrelor (Kengreal)	IV bolus 30 mcg/kg → 4 mcg/kg/min during PCI	IV, reversible, very short half-life (3–5 min) • for P2Y12-naive patients at PCI • transition to oral (clopidogrel/prasugrel given after stopping infusion; ticagrelor can be given during)

- **Prasugrel or ticagrelor preferred over clopidogrel** for ACS patients undergoing PCI (more potent; TRITON, PLATO) • ticagrelor also for medically managed ACS • **clopidogrel** if high bleeding risk, with fibrinolysis, or with oral anticoagulant.
- **Platelet transfusion** doesn't reverse ticagrelor well (reversible binding).

Duration of DAPT after ACS

- **Default: DAPT (aspirin + P2Y12 inhibitor) for 12 months** after ACS (PCI or medical therapy).
- **Not high bleeding risk:** after ≥ 1 month of DAPT, transition to **ticagrelor (or prasugrel) monotherapy** (stop aspirin) is reasonable to ↓ bleeding.
- **High bleeding risk** (PRECISE-DAPT ≥ 25, ARC-HBR): shorten DAPT to **1–3 months** then P2Y12 inhibitor monotherapy (or aspirin).
- **Extended > 12 months** in high ischemic, low bleeding risk (DAPT score ≥ 2): ticagrelor 60 mg BID, or clopidogrel/prasugrel.
- **De-escalation** (switch prasugrel/ticagrelor → clopidogrel) for bleeding or cost.
- **PPI** (pantoprazole) for patients on DAPT at high GI bleeding risk (history of GI bleed, age, steroids, anticoagulants).
- **Elective surgery:** delay ≥ 6 months after DES (≥ 1 month BMS) if possible; continue aspirin perioperatively.
- **Never stop DAPT prematurely** after stent placement (stent thrombosis → MI/death) without consulting the cardiologist.

ACS + atrial fibrillation (needs oral anticoagulant)

- **Triple therapy** (OAC + aspirin + clopidogrel) ↑ bleeding → keep as short as possible: **aspirin for ≤ 1 week (up to 30 days)** after PCI, then **OAC + clopidogrel** (dual therapy) up to **12 months** (6 months in CCD/high bleeding risk), then **OAC alone**.
- **DOAC preferred over warfarin** (stroke-prevention doses: apixaban 5 mg BID, rivaroxaban 15 mg daily [AF + PCI dose in PIONEER], dabigatran 150 mg BID, edoxaban 60 mg) • **clopidogrel is the P2Y12 of choice** (avoid prasugrel; ticagrelor in selected).

Parenteral anticoagulants in ACS

Agent	Dose	Notes
UFH	**60 units/kg IV bolus (max 4,000) → 12 units/kg/h (max 1,000 units/h)** • aPTT 1.5–2.5× control • during PCI: ACT-guided boluses	All strategies • preferred if CABG likely, severe CKD • HIT, bleeding • reversal protamine
Enoxaparin	**1 mg/kg SC q12h** (± 30 mg IV bolus) • **CrCl < 30: 1 mg/kg q24h** • STEMI with lysis age ≥ 75: no bolus, 0.75 mg/kg q12h	Invasive or conservative • avoid switching between UFH and enoxaparin • hold before CABG (12–24 h)
Bivalirudin	0.75 mg/kg IV bolus → 1.75 mg/kg/h during PCI (± continue 3–4 h post-PCI)	Direct thrombin inhibitor • **PCI only** • less bleeding • **HIT** patients • renal dose adjustment
Fondaparinux	2.5 mg SC daily	Conservative strategy/NSTE-ACS • **not as sole anticoagulant during PCI** (catheter thrombosis → add UFH) • CI CrCl < 30
GP IIb/IIIa inhibitors (eptifibatide, tirofiban)	Eptifibatide 180 mcg/kg bolus × 2 → 2 mcg/kg/min (CrCl < 50: 1 mcg/kg/min) • tirofiban 25 mcg/kg → 0.15 mcg/kg/min	**Bailout** during PCI (large thrombus, no-reflow) — not routine upstream • bleeding, **thrombocytopenia** • renal adjustment

- Anticoagulation is usually **stopped after successful PCI** (unless other indication: LV thrombus, AF, VTE).

8. Secondary Prevention After ACS (at discharge)

Therapy	Details
Aspirin	**81 mg daily indefinitely**
P2Y12 inhibitor	Ticagrelor or prasugrel (PCI) / clopidogrel — **12 months** default (see DAPT strategies)
High-intensity statin	**Atorvastatin 80 mg** / rosuvastatin 20–40 mg • lipid panel 4–8 weeks • if LDL ≥ 70 (or ≥ 55 very high risk) → **add ezetimibe** → **PCSK9 inhibitor** • goal ≥ 50% reduction and **LDL < 55 mg/dL** (2025 ACS guideline — reasonable)
β-blocker	Indicated with **LVEF ≤ 40%** (carvedilol, metoprolol succinate, bisoprolol) • reasonable for ~1 year in others; long-term benefit with preserved EF is uncertain
ACEi/ARB	**LVEF ≤ 40%**, HTN, diabetes, CKD (reasonable in all) • ARNI if HFrEF
MRA (eplerenone/spironolactone)	**LVEF ≤ 40% with HF symptoms or diabetes**, already on ACEi + β-blocker, SCr ≤ 2.5 (men)/2.0 (women), K ≤ 5.0 (EPHESUS)
SGLT2i	If HFrEF, diabetes, or CKD
SL nitroglycerin	PRN for chest pain + counseling
Others	**Cardiac rehabilitation** (referral before discharge) • smoking cessation • BP < 130/80 • diabetes management • influenza vaccine • weight management • screen for depression • **avoid NSAIDs** • sexual activity usually safe after 1 week if stable (no PDE-5 inhibitors within 24–48 h of nitrates)
Oral anticoagulant	LV thrombus (warfarin/DOAC 3 months), AF, VTE
ICD	If LVEF ≤ 35% **≥ 40 days after MI** (≥ 90 days after revascularization) on GDMT, NYHA II–III • (LVEF ≤ 30% NYHA I)
Colchicine	Low-dose colchicine early after MI is **not routinely recommended** (CLEAR SYNERGY showed no benefit) — may consider in CCD

Special ACS situations

- **Cocaine/methamphetamine-associated ACS: benzodiazepines** first, nitrates, CCBs • **avoid β-blockers** acutely (unopposed α-stimulation → vasospasm, ↑ BP).
- **RV infarction** (inferior STEMI, hypotension, clear lungs, JVD): **IV fluids**, **avoid nitrates and diuretics** (preload dependent), inotropes if needed.

- **Cardiogenic shock:** emergent revascularization, norepinephrine, inotropes (dobutamine/milrinone), mechanical circulatory support.
- **Women, elderly, CKD:** same evidence-based therapy with dose adjustments (renal dosing of enoxaparin, eptifibatide, bivalirudin).
- **Post-MI depression**: sertraline/escitalopram safe (avoid TCAs — arrhythmias).

9. Ischemic Heart Disease Drug Summary

Drug	Dose	Key ADRs	Key notes / interactions
Aspirin	ACS **162–325 mg chewed → 81 mg daily** • CCD 75–100 mg	GI bleeding, dyspepsia, bronchospasm (AERD), tinnitus (high dose)	≤ 100 mg with ticagrelor • ibuprofen interferes (take aspirin 30 min before or 8 h after) • Reye's (children)
Clopidogrel	**600 mg** (PCI) / 300 mg (lysis; none if > 75) → **75 mg**	Bleeding, TTP (rare), rash	**CYP2C19 prodrug** (poor metabolizers boxed) • **avoid omeprazole/esomeprazole** • hold 5 d pre-CABG
Prasugrel	**60 mg → 10 mg** (5 mg < 60 kg)	**Bleeding** (fatal)	**CI prior stroke/TIA** • not ≥ 75 y • PCI only • hold 7 d
Ticagrelor	**180 mg → 90 mg BID** (60 mg BID after 1 year)	**Dyspnea**, bleeding, bradyarrhythmia, ↑ uric acid	**Aspirin ≤ 100 mg** • CYP3A4 (avoid strong inhibitors/inducers) • hold 3–5 d • reversible
Cangrelor	30 mcg/kg → 4 mcg/kg/min IV	Bleeding, dyspnea	PCI, short half-life
UFH	ACS **60 units/kg (max 4,000) → 12 units/kg/h (max 1,000)**	Bleeding, **HIT**, hyperkalemia	aPTT 1.5–2.5× • protamine
Enoxaparin	**1 mg/kg SC q12h** (CrCl < 30 q24h) • STEMI lysis: 30 mg IV + 1 mg/kg (≥ 75: 0.75 mg/kg, no bolus)	Bleeding, HIT (less)	Renal adjust • spinal hematoma (boxed)
Bivalirudin	0.75 mg/kg → 1.75 mg/kg/h (PCI)	Bleeding	PCI, HIT • renal adjust
Fondaparinux	2.5 mg SC daily	Bleeding	Not alone for PCI • CI CrCl < 30
Eptifibatide / tirofiban	Eptifibatide 180 mcg/kg × 2 → 2 mcg/kg/min	Bleeding, **thrombocytopenia**	Bailout PCI • renal adjust
Tenecteplase	Weight-based single bolus (30–50 mg) • half-dose ≥ 75 y	**Bleeding, ICH**	STEMI if PCI > 120 min • door-to-needle 30 min
Alteplase	15 mg → 0.75 mg/kg (≤ 50) over 30 min → 0.5 mg/kg (≤ 35) over 60 min	**ICH**, bleeding, angioedema (ACEi)	Max 100 mg
Nitroglycerin SL	**0.4 mg q5 min × 3** (call 911 if not better 5 min after 1st)	**Headache**, hypotension, reflex tachycardia	**CI PDE-5i** (24/48 h), riociguat, RV infarct, SBP < 90 • glass bottle
Nitroglycerin IV	5–10 mcg/min → titrate (max 200)	Hypotension, headache, tolerance (24–48 h)	Ongoing ischemia, HTN, pulmonary edema
Isosorbide mononitrate ER	30–60 mg every morning (max 240)	Headache, hypotension	Once daily gives nitrate-free interval
Isosorbide dinitrate	10–40 mg TID (14-h gap)	Headache	Also with hydralazine in HFrEF
NTG patch	0.2–0.8 mg/h, on 12–14 h, off 10–12 h	Headache, skin irritation	Remove at night
Metoprolol	Tartrate 25–50 mg BID–QID (ACS) → succinate 25–200 mg daily	Bradycardia, hypotension, fatigue, HF worsening	Target HR 55–60 • taper to stop • CYP2D6 (↑ with fluoxetine, paroxetine, bupropion)

Drug	Dose	Key ADRs	Key notes / interactions
Carvedilol	3.125–25 mg BID	Hypotension (α1), dizziness	Post-MI LV dysfunction, HFrEF • with food
Amlodipine	5–10 mg daily	**Edema**, headache	Safe in HFrEF • simvastatin max 20 mg
Diltiazem / verapamil	Diltiazem ER 120–360 mg • verapamil ER 120–480 mg	Bradycardia, AV block, constipation (verapamil)	**Avoid HFrEF** • 3A4 inhibitors • vasospastic angina
Ranolazine	**500 → 1,000 mg BID**	Dizziness, constipation, **QT**	**CI strong 3A4 inhibitors/inducers, cirrhosis** • max 500 BID with diltiazem/verapamil • simvastatin ≤ 20, metformin ≤ 1,000 • ↑ digoxin
Atorvastatin	**80 mg** (ACS) / 40–80 mg	Myalgia, ↑ LFTs	Start early in ACS • 3A4 interactions
Lisinopril	2.5–5 mg → 10–40 mg daily	Cough, hyperkalemia, angioedema, ↑ SCr	Within 24 h (anterior MI, EF ≤ 40%) • CI pregnancy
Eplerenone	25 → 50 mg daily	**Hyperkalemia**	Post-MI EF ≤ 40% + HF/DM • CYP3A4 (CI strong inhibitors) • K ≤ 5.0
Colchicine	0.5 mg daily	GI, myopathy	CCD • CI strong 3A4/P-gp inhibitors, severe renal/hepatic disease
Rivaroxaban (vascular dose)	2.5 mg BID + aspirin	Bleeding	CCD/PAD high risk (COMPASS)
Morphine	2–4 mg IV	Hypotension, respiratory depression	Only refractory pain • delays oral P2Y12 effect

Exam Traps and Frequently Tested Points

1. **HR is the main determinant of myocardial O2 demand** • β-blockers ↓ demand • nitrates ↓ preload • CCBs ↓ afterload/vasospasm.
2. **Stable angina** = exertional, relieved by rest/NTG • **unstable angina** = rest/new/crescendo, troponin negative • **NSTEMI** = troponin positive, no ST elevation • **STEMI** = ST elevation.
3. **Variant (Prinzmetal) angina → CCBs + nitrates**, avoid nonselective β-blockers.
4. **Hold caffeine/theophylline 12–24 h before adenosine/regadenoson stress testing**.
5. **CCD event reduction:** aspirin 75–100 mg (clopidogrel if intolerant), high-intensity statin, BP < 130/80, SGLT2i/GLP-1 RA in diabetes, ACEi/ARB if HTN/DM/CKD/EF ≤ 40%.
6. **β-blocker long term only if** LVEF ≤ 50%, MI within 1 year, or other indication (angina, arrhythmia, HTN).
7. **Antianginal first-line: β-blocker or CCB** → add long-acting nitrate or ranolazine • avoid β-blocker + verapamil/diltiazem • avoid IR nifedipine.
8. **β-blocker target HR 55–60** • no ISA • never stop abruptly • avoid in cocaine ACS and vasospastic angina.
9. **Nitrate tolerance → 10–14 h nitrate-free interval** (ISMN ER once in AM, patch off at night, ISDN TID with 14-h gap).
10. **Nitrates + PDE-5 inhibitors = contraindicated** (sildenafil/vardenafil 24 h, tadalafil 48 h) • also riociguat • RV infarction, SBP < 90.
11. **SL NTG 0.4 mg:** sit, repeat q5 min up to 3 doses, **call 911 if no improvement 5 min after the first dose** • original glass container.
12. **Ranolazine:** no effect on HR/BP • QT • **CI strong CYP3A4 inhibitors/inducers and cirrhosis** • simvastatin ≤ 20, metformin ≤ 1,000 • max 1,000 mg BID.
13. **ACS early therapy:** aspirin 162–325 chewed, NTG, O2 only if SpO2 < 90%, P2Y12, anticoagulant, high-intensity statin, oral β-blocker within 24 h, ACEi within 24 h (anterior MI/EF ≤ 40%).

14. **β-blocker contraindications in ACS:** HF, low output, cardiogenic shock risk, PR > 0.24 s, 2nd/3rd-degree block, active asthma • no IV β-blocker if shock risk.
15. **Avoid NSAIDs** after MI • morphine only for refractory pain (delays P2Y12 absorption).
16. **STEMI:** primary PCI if FMC-to-device ≤ 90 min (≤ 120 with transfer) • otherwise **fibrinolysis within 30 min** (symptoms < 12 h) → transfer, angiography 3–24 h.
17. **Fibrinolytics:** tenecteplase single weight-based bolus (half dose ≥ 75) • alteplase 15 mg + 0.75 mg/kg + 0.5 mg/kg • **never in NSTE-ACS** • know absolute contraindications.
18. **With fibrinolysis:** clopidogrel 300 mg (no load if > 75 → 75 mg) + enoxaparin (≥ 75: no bolus, 0.75 mg/kg).
19. **NSTE-ACS invasive timing:** immediate < 2 h (refractory angina, instability) • early < 24 h (GRACE > 140).
20. **Clopidogrel:** CYP2C19 prodrug, poor metabolizers ↓ effect, avoid omeprazole/esomeprazole (use pantoprazole) • 600 mg load for PCI.
21. **Prasugrel:** CI with prior stroke/TIA, avoid ≥ 75 years, 5 mg if < 60 kg, PCI only • **ticagrelor:** aspirin ≤ 100 mg, dyspnea, bradycardia, ↑ uric acid, CYP3A4.
22. **Hold before CABG:** clopidogrel 5 days, ticagrelor 3–5 days, prasugrel 7 days.
23. **DAPT 12 months after ACS** (shorten to 1–3 months if high bleeding risk; P2Y12 monotherapy after ≥ 1 month reasonable) • elective PCI in CCD: 6 months clopidogrel + aspirin.
24. **AF + PCI:** triple therapy ≤ 1 week (up to 30 days) → OAC (DOAC) + clopidogrel to 12 months → OAC alone • clopidogrel is the P2Y12 of choice.
25. **Anticoagulants:** UFH 60 units/kg + 12 units/kg/h • enoxaparin 1 mg/kg q12h • bivalirudin (PCI, HIT) • **fondaparinux not alone for PCI** • GP IIb/IIIa bailout only.
26. **Discharge after MI:** aspirin, P2Y12, high-intensity statin (LDL < 55–70; add ezetimibe → PCSK9i), β-blocker (EF ≤ 40%), ACEi/ARB, MRA (EF ≤ 40% + HF/DM), SL NTG, cardiac rehab.
27. **ICD** if LVEF ≤ 35% ≥ 40 days post-MI (≥ 90 days post-revascularization) on GDMT.
28. **Cocaine ACS:** benzodiazepines + nitrates/CCB, no β-blockers • **RV infarct:** fluids, avoid nitrates/diuretics.
29. **Ibuprofen blocks aspirin's antiplatelet effect** → aspirin 30 min before or 8 h after ibuprofen.

Note: doses and recommendations follow the ACC/AHA 2023 Chronic Coronary Disease guideline, the 2025 ACC/AHA ACS guideline, and ESC guidance. If your course slides or reference use slightly different numbers, **use your course numbers for the exam**.